THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

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ANSWER KEY & MARKING SCHEME SET 1

Chapter 2: **Polynomials** | Class 10 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** 6	Substitute $x = 4$: $p(4) = 6$
2	**(d)** 4	$2x - 8 = 0 \Rightarrow x = 4$
3	**(d)** 5	Sum of zeroes $= -b/a$
4	**(b)** 6	Product of zeroes $= c/a$
5	**(d)** 2	A polynomial of degree n has at most n zeroes
6	**(b)** 2	$p(2) = 0 = 0$, while the other values do not make $p(x)$ zero
7	**(a)** 16	Substitute $x = 3$: $p(3) = 16$
8	**(c)** -3	$3x + 9 = 0 \Rightarrow x = -3$
9	**(b)** 4	Sum of zeroes $= -b/a$
10	**(c)** -5	Product of zeroes $= c/a$

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	$p(3) = 10$ (2).	2
12	$3x - 12 = 0$ (1) so $x = 4$ (1).	1+1
13	Factorise: $(x - 2)(x - 5)$ (1); zeroes 2 and 5 (1).	1+1
14	$p(-2) = 0$ (2).	2

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Factors $(x - 3)(x - 5)$ (1); zeroes $3, 5$ (1); sum $8 = -b/a$ verified (1).	1+1+1
16	$p(x) = k[x^2 - (\text{sum})x + \text{product}]$ (1); $x^2 - (5)x + (6)$ (1); one such polynomial is $x^2 - 5x + 6$ (1).	1+1+1
17	Put $x = 1$: $p(1) = 0$ (2) $= 0$, hence a zero (1).	2+1
	OR: Factors $(x + 2)(x - 6)$ (1); zeroes $-2, 6$ (1); sum $4 = -b/a$ verified (1).	1+1+1
18	$p(x) = k[x^2 - (\text{sum})x + \text{product}]$ (1); $x^2 - (-3)x + (2)$ (1); one such polynomial is $x^2 + 3x + 2$ (1).	1+1+1
	OR: Put $x = 2$: $p(2) = 0$ (2) $= 0$, hence a zero (1).	2+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Zeroes 2, 3 (2); sum $= 5 = -\frac{b}{a}$ (1.5); product $= 6 = \frac{c}{a}$ (1.5).	2+1.5+1.5
	OR: Sum $= 6 = -\frac{b}{a}$ (2); sum of products in pairs $= 11 = \frac{c}{a}$ (2); product $= 6 = -\frac{d}{a}$ (1).	2+2+1
20	Zeroes -4, 1 (2); sum $= -3 = -\frac{b}{a}$ (1.5); product $= -4 = \frac{c}{a}$ (1.5).	2+1.5+1.5
	OR: Sum $= 5 = -\frac{b}{a}$ (2); sum of products in pairs $= 2 = \frac{c}{a}$ (2); product $= -8 = -\frac{d}{a}$ (1).	2+2+1

Mistakes to avoid:

- Writing sum of zeros as $\frac{c}{a}$ and product as $-\frac{b}{a}$; it is sum $= -\frac{b}{a}$ and product $= \frac{c}{a}$.
- Forgetting the sign: for $x^2 - 8x + 15$ the sum of zeros is $+8$, not -8.
- Counting the zeros of $y = p(x)$ as the y-intercept; zeros are where the graph meets the *x*-axis.
- Counting a parabola that only touches the x-axis as having no zeros or two different zeros; it has one repeated zero.
- Writing the polynomial from its zeros as $x^2 + (\alpha + \beta)x + \alpha\beta$ instead of $x^2 - (\alpha + \beta)x + \alpha\beta$.
- Claiming a quadratic polynomial always has two real zeros (e.g. $x^2 + 1$ has none).

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

ANSWER KEY & MARKING SCHEME SET 2

Chapter 2: **Polynomials** | Class 10 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** 3	A polynomial of degree n has at most n zeroes
2	**(d)** 3	$p(3) = 0 = 0$, while the other values do not make $p(x)$ zero
3	**(b)** 16	Substitute $x = 5$: $p(5) = 16$
4	**(d)** 4	$5x - 20 = 0 \Rightarrow x = 4$
5	**(a)** 10	Sum of zeroes $= -b/a$
6	**(d)** 24	Product of zeroes $= c/a$
7	**(a)** 4	A polynomial of degree n has at most n zeroes
8	**(d)** 2	$p(2) = 0 = 0$, while the other values do not make $p(x)$ zero
9	**(a)** 9	Substitute $x = 2$: $p(2) = 9$
10	**(d)** -3	$4x + 12 = 0 \Rightarrow x = -3$

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	$5x + 15 = 0$ (1) so $x = -3$ (1).	1+1
12	Factorise: $(x + 3)(x - 4)$ (1); zeroes -3 and 4 (1).	1+1
13	$p(2) = 8$ (2).	2
14	$2x - 9 = 0$ (1) so $x = \frac{9}{2}$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Factors $(x - 4)(x - 7)$ (1); zeroes $4, 7$ (1); sum $11 = -b/a$ verified (1).	1+1+1
16	$p(x) = k[x^2 - (\text{sum})x + \text{product}]$ (1); $x^2 - (7)x + (10)$ (1); one such polynomial is $x^2 - 7x + 10$ (1).	1+1+1
17	Put $x = 3$: $p(3) = 0$ (2) $= 0$, hence a zero (1).	2+1
	OR: Factors $(x + 5)(x - 2)$ (1); zeroes $-5, 2$ (1); sum $-3 = -b/a$ verified (1).	1+1+1
18	$p(x) = k[x^2 - (\text{sum})x + \text{product}]$ (1); $x^2 - (0)x + (-9)$ (1); one such polynomial is $x^2 - 9$ (1).	1+1+1
	OR: Put $x = -1$: $p(-1) = 0$ (2) $= 0$, hence a zero (1).	2+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Zeroes $5, 6$ (2); sum $= 11 = -\frac{b}{a}$ (1.5); product $= 30 = \frac{c}{a}$ (1.5).	2+1.5+1.5
	OR: Sum $= 3 = -\frac{b}{a}$ (2); sum of products in pairs $= -4 = \frac{c}{a}$ (2); product $= -12 = -\frac{d}{a}$ (1).	2+2+1
20	Zeroes $-2, 7$ (2); sum $= 5 = -\frac{b}{a}$ (1.5); product $= -14 = \frac{c}{a}$ (1.5).	2+1.5+1.5
	OR: Sum $= 3 = -\frac{b}{a}$ (2); sum of products in pairs $= -13 = \frac{c}{a}$ (2); product $= -15 = -\frac{d}{a}$ (1).	2+2+1

Mistakes to avoid:

- Writing sum of zeros as $\frac{c}{a}$ and product as $-\frac{b}{a}$; it is sum $= -\frac{b}{a}$ and product $= \frac{c}{a}$.
- Forgetting the sign: for $x^2 - 8x + 15$ the sum of zeros is $+8$, not -8.
- Counting the zeros of $y = p(x)$ as the y-intercept; zeros are where the graph meets the *x*-axis.
- Counting a parabola that only touches the x-axis as having no zeros or two different zeros; it has one repeated zero.
- Writing the polynomial from its zeros as $x^2 + (\alpha + \beta)x + \alpha\beta$ instead of $x^2 - (\alpha + \beta)x + \alpha\beta$.
- Claiming a quadratic polynomial always has two real zeros (e.g. $x^2 + 1$ has none).

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

ANSWER KEY & MARKING SCHEME SET 3

Chapter 2: **Polynomials** | Class 10 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** -5	Sum of zeroes $= -b/a$
2	**(b)** 6	Product of zeroes $= c/a$
3	**(c)** 5	A polynomial of degree n has at most n zeroes
4	**(c)** 4	$p(4) = 0 = 0$, while the other values do not make $p(x)$ zero
5	**(c)** 9	Substitute $x = 1$: $p(1) = 9$
6	**(b)** 3	$7x - 21 = 0 \Rightarrow x = 3$
7	**(b)** 8	Sum of zeroes $= -b/a$
8	**(c)** 7	Product of zeroes $= c/a$
9	**(a)** -4	$p(-4) = 0 = 0$, while the other values do not make $p(x)$ zero
10	**(a)** 18	Substitute $x = 5$: $p(5) = 18$

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Factorise: $(x-1)(x-6)$ (1); zeroes 1 and 6 (1).	1+1
12	$p(5) = 21$ (2).	2
13	$4x + 6 = 0$ (1) so $x = -\frac{3}{2}$ (1).	1+1
14	Factorise: $(x+2)(x+7)$ (1); zeroes -2 and -7 (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Factors $(x+3)(x+4)$ (1); zeroes $-3, -4$ (1); sum $-7 = -b/a$ verified (1).	1+1+1
16	$p(x) = k[x^2 - (\text{sum})x + \text{product}]$ (1); $x^2 - (4)x + (-21)$ (1); one such polynomial is $x^2 - 4x - 21$ (1).	1+1+1
17	Put $x = 2$: $p(2) = 0$ (2) $= 0$, hence a zero (1).	2+1
	OR: Factors $(x-1)(x-9)$ (1); zeroes $1, 9$ (1); sum $10 = -b/a$ verified (1).	1+1+1
18	$p(x) = k[x^2 - (\text{sum})x + \text{product}]$ (1); $x^2 - (-8)x + (15)$ (1); one such polynomial is $x^2 + 8x + 15$ (1).	1+1+1
	OR: Put $x = -2$: $p(-2) = 0$ (2) $= 0$, hence a zero (1).	2+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Zeroes $\frac{1}{2}$, 3 (2); sum $= \frac{7}{2} = -\frac{b}{a}$ (1.5); product $= \frac{3}{2} = \frac{c}{a}$ (1.5).	2+1.5+1.5
	OR: Sum $= 12 = -\frac{b}{a}$ (2); sum of products in pairs $= 44 = \frac{c}{a}$ (2); product $= 48 = -\frac{d}{a}$ (1).	2+2+1
20	Zeroes $\frac{2}{3}$, -1 (2); sum $= -\frac{1}{3} = -\frac{b}{a}$ (1.5); product $= -\frac{2}{3} = \frac{c}{a}$ (1.5).	2+1.5+1.5
	OR: Sum $= 0 = -\frac{b}{a}$ (2); sum of products in pairs $= -7 = \frac{c}{a}$ (2); product $= 6 = -\frac{d}{a}$ (1).	2+2+1

Mistakes to avoid:

- Writing sum of zeros as $\frac{c}{a}$ and product as $-\frac{b}{a}$; it is sum $= -\frac{b}{a}$ and product $= \frac{c}{a}$.
- Forgetting the sign: for $x^2 - 8x + 15$ the sum of zeros is $+8$, not -8.
- Counting the zeros of $y = p(x)$ as the y-intercept; zeros are where the graph meets the *x*-axis.
- Counting a parabola that only touches the x-axis as having no zeros or two different zeros; it has one repeated zero.
- Writing the polynomial from its zeros as $x^2 + (\alpha+\beta)x + \alpha\beta$ instead of $x^2 - (\alpha+\beta)x + \alpha\beta$.
- Claiming a quadratic polynomial always has two real zeros (e.g. $x^2 + 1$ has none).

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.